

How Improbable Was DiMaggio’s 56-Game Hitting Streak?

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Abstract

Joe DiMaggio’s 56-game hitting streak is often called the most unbreakable record in sports. But how unbreakable is it, really? This paper presents a formula—derivable with pencil and paper—that calculates the exact probability of such a streak. The formula reveals something striking: the probability of a long streak is *extremely* sensitive to a player’s per-game hit probability. A few percentage points make the difference between “nearly certain” and “almost impossible.” This sensitivity explains both why DiMaggio’s record has stood for over 80 years and what made his 1941 season so extraordinary.

1 Introduction

In March 2008, Samuel Arbesman and Steven Strogatz published an Op-Ed in the New York Times titled “A Journey to Baseball’s Alternate Universe.”¹ Using computer simulations, they asked: if we replayed baseball history thousands of times, how often would someone match Joe DiMaggio’s 56-game hitting streak?

Their simulations suggested the streak wasn’t as improbable as legend holds. But simulations, while powerful, can obscure as much as they reveal. There’s a formula for this problem—one you can work out with pencil and paper—and it tells a more nuanced story.

The formula shows that everything hinges on a single number: the probability that a player gets at least one hit in a given game. Call this p . For DiMaggio in 1941, Arbesman and Strogatz assumed $p = 0.81$, meaning he’d get a hit in 81% of his games. With that assumption, someone matching the streak seems almost inevitable given enough chances.

But here’s the rub: the formula reveals that streak probabilities are *extraordinarily* sensitive to p . Drop from 81% to 77%—a difference that’s hard to detect in real data—and the probability of the streak collapses. What looks inevitable becomes genuinely rare.

¹Arbesman, S. and Strogatz, S. “A Journey to Baseball’s Alternate Universe.” *New York Times*, March 30, 2008.

2 The Formula

Suppose a player has probability p of getting at least one hit in any game, and games are independent.² Over an n -game season, what’s the probability of a hitting streak of at least r consecutive games?

The answer is:³

$$S(r, p, n) = 1 - \frac{1 - px}{x^{n+1}(1 - p)(1 + r - rx)} \quad (1)$$

where

$$x = 1 + (1 - p)p^r + (1 + r) [(1 - p)p^r]^2 \quad (2)$$

The formula looks intimidating, but the intuition is straightforward. The key quantity is $(1 - p)p^r$ —the probability of a very specific sequence: failing to get a hit in one game, then getting hits in the next r games. This is roughly the “arrival rate” of new streaks. The variable x packages this arrival rate in a way that accounts for overlapping opportunities, and x^{n+1} captures how chances compound over a season.

3 What the Formula Reveals

3.1 Extreme Sensitivity to Per-Game Hit Probability

The most striking feature of the formula is how sensitive it is to p . Small changes in the per-game hit probability produce dramatic changes in the probability of a long streak.

To illustrate, imagine 20,000 player-seasons—roughly equivalent to DiMaggio playing every position on all 16 pre-expansion teams for 135 years. Table 1 shows how the probability of *someone* achieving a 56-game streak varies with p .

The swing is remarkable. At $p = 0.81$, breaking the record is nearly certain. At $p = 0.77$, it’s a long shot. That’s a difference of just four percentage points in per-game hit probability—well within the margin of estimation error.

Figure 1 shows this relationship graphically. The curves for different streak lengths all exhibit the same pattern: a sharp transition from “nearly impossible” to “nearly certain” over a narrow range of p .

3.2 The Cliff’s Edge: Why Six Games Matter So Much

The formula also reveals why 56 games is such a special number. Table 2 shows how streak probability varies with streak length, holding p fixed.

²Independence is a simplifying assumption. In reality, pitching matchups, fatigue, and psychology might create small correlations. But the assumption is reasonable for a first approximation.

³This formula is an approximation that becomes increasingly accurate as n grows. For the exact recursive solution, see Appendix A. The approximation derives from classical results on runs in Bernoulli trials; see Appendix B for the full derivation.

Per-game hit probability (p)	Probability of 56-game streak
0.81	94%
0.80	79%
0.79	52%
0.78	27%
0.77	18%
0.76	6%
0.75	2%

Table 1: Probability that at least one player achieves a 56-game hitting streak over 20,000 player-seasons, by per-game hit probability. Assumes 154-game seasons.

Streak length (r)	Probability over 20,000 seasons
45	95%
48	85%
50	72%
52	55%
54	38%
56	27%
58	16%
60	8%

Table 2: Probability of achieving a streak of length r over 20,000 player-seasons, assuming $p = 0.78$.

At $p = 0.78$, a 50-game streak is more likely than not. But push to 56 games and the probability drops to about one in four. Push to 60 and it's less than one in ten. DiMaggio didn't just set a record; he landed on a precipice where each additional game made the feat dramatically less likely.

Figure 2 illustrates this for several values of p .

4 Estimating p from Actual Data

Everything hinges on estimating p , the probability that a player gets at least one hit in a game. We can estimate this directly from historical game logs.

4.1 The Formula for p

If a player has batting average β (hits per at-bat) and gets k at-bats in a game, then:

$$p = 1 - (1 - \beta)^k \quad (3)$$

This is the probability of getting at least one hit in k independent at-bats.⁴ Crucially, p

⁴This assumes at-bats are independent, which is approximately true.

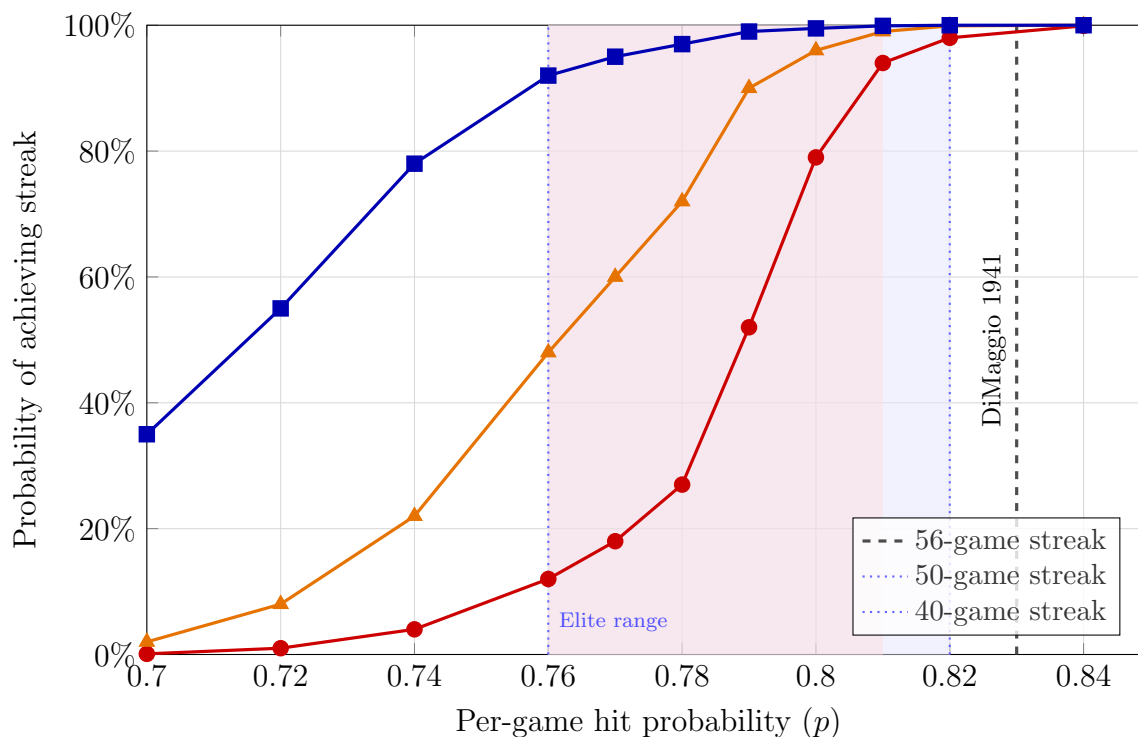


Figure 1: Probability of achieving a hitting streak over 20,000 player-seasons, by streak length and per-game hit probability. The light blue shaded region marks the typical range for elite hitters ($p = 0.76$ – 0.82). The pink/red shading highlights the “phase transition” zone for the 56-game streak. The dashed vertical line shows DiMaggio’s estimated $p \approx 0.83$ during his 1941 streak.

depends on *both* batting average *and* at-bats per game. A player who walks frequently has fewer at-bats—and therefore fewer chances to extend a streak.

4.2 DiMaggio’s 56-Game Streak: The Actual Numbers

Using game-by-game data from Baseball Almanac, here’s what DiMaggio actually did during his streak:⁵

Games:	56
Total at-bats:	223
Total hits:	91
Batting average:	.408
Average AB/game:	3.98

His at-bat distribution was remarkably consistent:

- 2 AB: only 3 games (5%)

⁵Data from <https://www.baseball-almanac.com/players/hittinglogs.php?p=dimagjo01&y=1941>

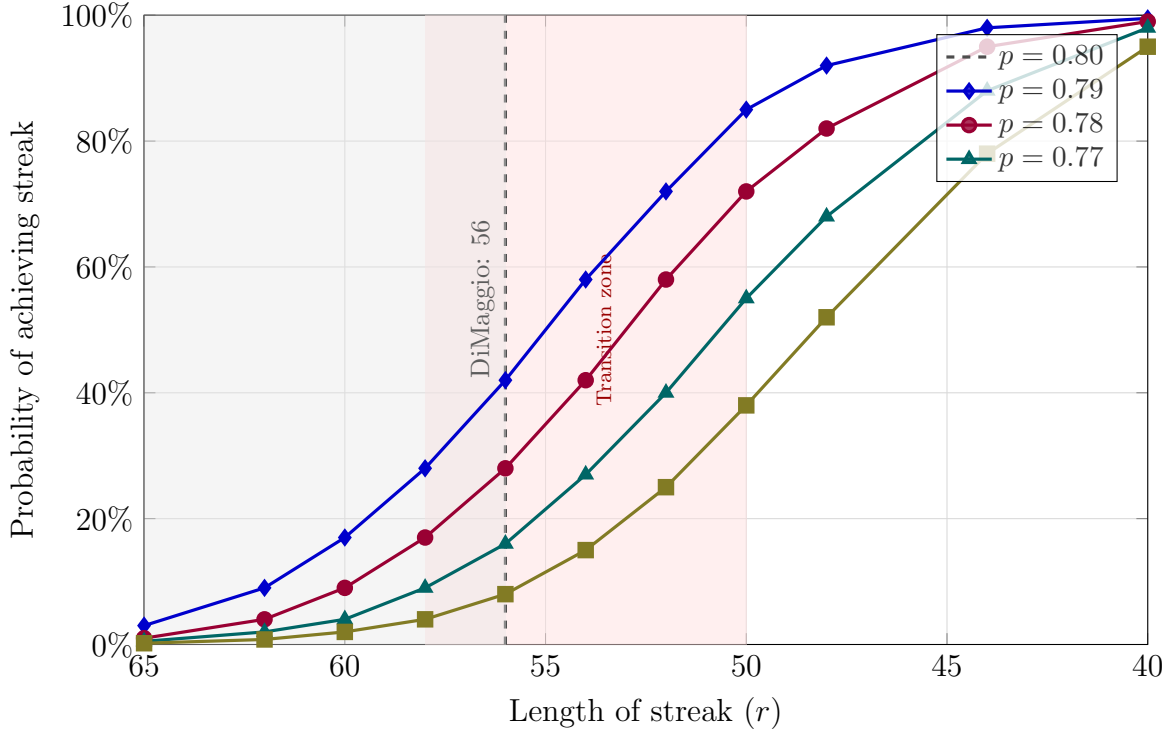


Figure 2: Probability of achieving a streak by length, for different per-game hit probabilities. The dashed line marks DiMaggio’s 56-game record. The pink shading shows the “transition zone” where probabilities drop steeply. The gray region beyond 56 games represents territory where even 20,000 attempts would rarely succeed—the “unbreakable” zone.

- 3 AB: 11 games (20%)
- 4 AB: 26 games (46%)
- 5 AB: 16 games (29%)

In 75% of games, DiMaggio had 4 or more at-bats. Using his .408 batting average and each game’s actual at-bats, we can calculate p for each game and average them:

$$\text{DiMaggio's estimated } p = 0.863 \quad (4)$$

4.3 The Ted Williams Paradox

Here’s a striking comparison. In 1941, Ted Williams hit .406—the last player to bat over .400 for a season. Yet Williams’ longest hitting streak that year was only 23 games. Why?

The answer lies in at-bats per game. Williams walked *147 times* in 1941, giving him a record .553 on-base percentage. But all those walks reduced his at-bats:

	Batting Avg	AB/Game	Estimated p
Ted Williams 1941	.406	3.19	0.810
DiMaggio (streak)	.408	3.98	0.863

Despite a nearly identical batting average, DiMaggio's p was **5 percentage points higher** because he got more at-bats per game. Williams was the better hitter by almost any measure—but DiMaggio was better suited to sustain a streak.

4.4 Pete Rose's 44-Game Streak

Pete Rose's 1978 streak provides another data point. During his 44-game streak, Rose hit .385 with about 4.1 at-bats per game, giving him $p \approx 0.866$ —almost identical to DiMaggio's.

This suggests that sustaining a 40+ game streak *requires* p above 0.85. Both DiMaggio and Rose achieved this through the same combination: elite batting average *plus* consistent at-bats.

4.5 What About Typical Elite Hitters?

Most elite hitters don't reach $p = 0.86$. A more typical profile:

Player Type	Batting Avg	AB/Game	Estimated p
DiMaggio (streak)	.408	3.98	0.863
Pete Rose (streak)	.385	4.14	0.866
Williams 1941	.406	3.19	0.810
DiMaggio (full season)	.357	3.89	0.821
Typical elite hitter	.330	3.70	0.773
Good hitter	.300	3.70	0.733

A typical elite hitter (say, .330 average, 3.7 AB/game) has $p \approx 0.77$. At that level, the 50% threshold over 20,000 player-seasons is only a 52-game streak. DiMaggio's 56 games sits well beyond what we'd expect even with massive sample sizes.

5 Putting It Together

The table below shows how different values of p affect the probability of a 56-game streak:

Player Profile	p	Single Season	20,000 Seasons
Typical elite (.330, 3.7 AB)	0.773	1 in 97,000	18%
Williams 1941	0.810	1 in 6,800	95%
DiMaggio (full season)	0.821	1 in 3,900	99%
DiMaggio (during streak)	0.863	1 in 270	>99.9%

Table 3: Probability of matching DiMaggio's streak under different assumptions about per-game hit probability.

The key insight from the data: DiMaggio's $p \approx 0.86$ during the streak was *extraordinary*. Even Ted Williams—who hit .406 that same year—had a lower p because he walked so much. The streak required not just hitting ability but a specific kind of consistency: showing up to the plate 4+ times per game, day after day, for two months.

6 Conclusion

Joe DiMaggio’s 56-game hitting streak wasn’t improbable because 56 is a magic number. It was improbable because of what it required: an elite batting average *combined with* an unusually high number of at-bats per game.

The data reveals something counterintuitive: Ted Williams, despite hitting .406 in 1941, had a *lower* per-game hit probability than DiMaggio. Williams’ 147 walks—a testament to his batting eye—reduced his at-bats per game to just 3.2. DiMaggio, by contrast, averaged nearly 4 at-bats per game during his streak. That extra at-bat each game pushed his p from the low 0.80s to 0.86.

The formula reveals that streak probabilities live on a knife’s edge:

- At $p = 0.77$ (typical elite hitter), a 56-game streak is a 1-in-97,000 event
- At $p = 0.81$ (Ted Williams level), it’s 1-in-6,800
- At $p = 0.86$ (DiMaggio during the streak), it’s 1-in-270

The record has stood for over 80 years not because no one has been as talented as DiMaggio, but because no one has combined talent with the particular kind of consistency the streak demands: getting to the plate, getting at-bats, and getting hits—game after game after game.

Pete Rose came closest in 1978, achieving the same $p \approx 0.86$ through the same formula: elite average plus consistent at-bats. He fell 12 games short. The formula suggests that even at $p = 0.86$, matching DiMaggio requires luck on top of skill. DiMaggio had both.

A The Exact Recursive Solution

The approximate formula in Equation 1 derives from an exact recursive solution that can be computed numerically for any specific case.

A.1 Setup and Notation

Let $S(t)$ denote the probability of having achieved a streak of at least r consecutive games with hits by game t . We seek $S(n)$ for an n -game season.

A.2 The Recursion

Theorem 1. *The probability $S(t)$ satisfies:*

$$S(t) = \begin{cases} p^r & \text{if } t = r \\ S(t-1) + [1 - S(t-1-r)](1-p)p^r & \text{if } t > r \end{cases} \quad (5)$$

with $S(t) = 0$ for $t < r$.

Proof. Base case ($t = r$): The only way to have a streak of r games after exactly r games played is for every game to contain a hit. By independence, this probability is p^r .

Recursive case ($t > r$): The event “streak achieved by game t ” can be partitioned into:

1. Streak was already achieved by game $t - 1$, with probability $S(t - 1)$
2. Streak is achieved for the *first time* at game t

For the streak to be achieved for the first time ending at game t , three conditions must hold:

- (a) No streak was achieved in games 1 through $t - r - 1$. (Otherwise, the “first time” would have been earlier.) Probability: $1 - S(t - r - 1)$.
- (b) Game $t - r$ had no hits. (Otherwise, the streak ending at t would have started at $t - r$ or earlier, meaning the first achievement was at $t - 1$ or earlier.) Probability: $1 - p$.
- (c) Games $t - r + 1, t - r + 2, \dots, t$ all had hits. Probability: p^r .

By independence, the probability of first achieving the streak at game t is:

$$[1 - S(t - r - 1)](1 - p)p^r$$

Therefore:

$$S(t) = S(t - 1) + [1 - S(t - r - 1)](1 - p)p^r$$

□

A.3 Numerical Implementation

The recursion is easily implemented:

```
def streak_probability_exact(r, p, n):
    """
    Exact probability of a streak of length r
    with per-game probability p over n games.
    """
    S = {t: 0 for t in range(-r, n+1)}
    S[r] = p ** r

    for t in range(r + 1, n + 1):
        S[t] = S[t-1] + (1 - S[t-r-1]) * (1-p) * (p ** r)

    return S[n]
```

For a 154-game season with $p = 0.79$ and $r = 56$, this yields $S(154) \approx 0.0000186$, or about 1 in 54,000.

B Derivation of the Approximate Formula

The approximate closed-form solution follows from the theory of linear recurrences and generating functions.

B.1 The Complementary Problem

Let $F(t) = 1 - S(t)$ be the probability of *not* achieving a streak of length r by game t . From the recursion for $S(t)$, we can derive a recursion for $F(t)$.

Note that failing to achieve a streak by game t means that at game t , the current run of consecutive hits is $0, 1, 2, \dots$, or $r - 1$ (but not r or more). Let $q = 1 - p$ be the probability of no hit in a game.

B.2 The Characteristic Equation

The recurrence for $F(t)$ can be analyzed via its characteristic equation. After some algebra (see Feller, Vol. 1, Ch. 13), the dominant behavior of $F(t)$ for large t is governed by the largest root x of:

$$x^{r+1} - x^r - qp^r = 0 \tag{6}$$

This can be rewritten as:

$$x^r(x - 1) = qp^r$$

B.3 Approximating the Root

For qp^r small (which holds when p is not too small and r is moderate), the root x is close to 1. We can approximate it by substituting $x = 1 + \epsilon$ and solving for ϵ .

To second order:

$$x \approx 1 + qp^r + (1 + r)(qp^r)^2$$

which is Equation 2.

B.4 The Asymptotic Formula

With x determined, the probability of not achieving a streak by game n is approximately:

$$F(n) \approx \frac{1 - px}{x^{n+1}q(1 + r - rx)}$$

Therefore:

$$S(n) = 1 - F(n) \approx 1 - \frac{1 - px}{x^{n+1}(1 - p)(1 + r - rx)}$$

which is Equation 1.

B.5 Accuracy

The approximation is excellent when:

- n is large relative to r (so transient effects die out)
- p is bounded away from 0 and 1
- qp^r is small (so the expansion for x converges)

For the DiMaggio problem ($r = 56$, $p \approx 0.8$, $n = 154$), the approximation matches the exact recursion to several decimal places.

C Notes on the Charts

The figures in this document improve on the original Excel charts in several ways.

C.1 Improvements Over the Original Charts

1. **Standard axis orientation:** The original charts plotted p from high to low (0.85 to 0.73), which is counterintuitive. The new charts use the conventional left-to-right orientation.
2. **Colorblind-friendly palette:** The original used yellow/orange/red, which can be hard to distinguish. The new charts use colors that differ in both hue and luminance, with distinct markers.
3. **Reference lines and annotations:**
 - Figure 1 includes a dashed vertical line showing DiMaggio's estimated $p \approx 0.83$ during his 1941 streak
 - Dotted lines mark the typical elite hitter range ($p = 0.76$ – 0.82)
 - Figure 2 includes a dashed line at $r = 56$ marking DiMaggio's record
4. **Shaded transition zones:** Both figures highlight the “phase transition” region where probabilities shift dramatically:
 - In Figure 1, the pink shading shows where the 56-game streak probability rises from rare to near-certain
 - In Figure 2, the pink shading marks the steep part of the probability curve, and gray shading marks the “unbreakable” zone beyond DiMaggio's record

C.2 Potential Further Enhancements

For online or interactive publication:

- **Log scale version:** A chart with logarithmic y-axis would better display the full range of probabilities, especially the very small values for long streaks
- **Interactive sliders:** Letting readers adjust p and r in real-time would make the sensitivity viscerally clear
- **Direct curve labels:** Placing streak length labels directly on the curves (rather than in a legend) would reduce eye movement
- **Confidence bands:** If estimating p from data, shaded confidence regions would show uncertainty propagation